

## TROWBRIDGE, ON, Canada

WMO: 714611

Lat: 48.292N

Lon: 88.876W

Elev: 715

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 88-00

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -12.3      | -7.7 | -23.3                         | 1.6 | -6.0 | -19.3 | 2.0 | -4.3 | 45.5                  | 27.9 | 42.4 | 20.5 | 19.1                     | 310  | 0.877 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.7                    | 72.0            | 59.2 | 69.1 | 59.0 | 66.7 | 58.6 | 63.6                | 67.8 | 62.0 | 66.3 | 60.6 | 65.0 | 14.4                 | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.1                            | 85.6 | 65.6 | 60.2 | 79.8 | 64.4 | 58.5 | 75.1 | 63.4 | 29.4          | 68.1 | 28.2 | 66.3 | 27.2 | 65.0 | 70.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 40.3          | 35.6 | 31.2 | -18.2                      | 78.7 | 5.3                | 4.4 | -22.0                                              | 81.8 | -25.0      | 84.4 | -28.0      | 86.9 | -31.8      | 90.1 |    |    |
| (5)               |      |      | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 38.8   | 14.9  | 16.6  | 25.9  | 36.5  | 46.0  | 52.3  | 58.3  | 61.2  | 55.2  | 43.5  | 32.2  | 21.3  |  |  |
|                                                                        | DBStd       | 17.82  | 12.00 | 12.20 | 9.24  | 6.25  | 5.42  | 5.24  | 5.35  | 4.36  | 6.07  | 6.52  | 8.14  | 12.61 |  |  |
|                                                                        | HDD50       | 5019   | 1090  | 934   | 746   | 407   | 146   | 31    | 3     | 0     | 22    | 217   | 533   | 890   |  |  |
|                                                                        | HDD65       | 9595   | 1555  | 1354  | 1211  | 856   | 588   | 381   | 217   | 133   | 295   | 667   | 983   | 1355  |  |  |
|                                                                        | CDD50       | 928    | 0     | 0     | 0     | 1     | 23    | 101   | 260   | 348   | 180   | 15    | 0     | 0     |  |  |
|                                                                        | CDD65       | 27     | 0     | 0     | 0     | 0     | 0     | 1     | 9     | 15    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 52     | 0     | 0     | 0     | 0     | 1     | 8     | 20    | 19    | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 3      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 15.3   | 18.0  | 17.2  | 14.9  | 14.0  | 13.1  | 12.9  | 12.5  | 11.7  | 15.1  | 17.1  | 18.9  | 18.8  |  |  |
|                                                                        | PrecAvg     | 34.6   | 1.9   | 1.3   | 1.7   | 2.6   | 3.5   | 3.9   | 3.8   | 3.5   | 4.1   | 3.6   | 2.7   | 2.1   |  |  |
| (15) Precipitation                                                     | PrecMax     | 41.3   | 4.4   | 3.3   | 3.3   | 7.7   | 8.1   | 8.0   | 6.7   | 5.4   | 8.5   | 6.6   | 5.2   | 3.7   |  |  |
|                                                                        | PrecMin     | 25.4   | 0.4   | 0.4   | 0.4   | 0.7   | 1.9   | 1.8   | 2.2   | 1.7   | 1.7   | 1.3   | 1.0   | 1.0   |  |  |
|                                                                        | PrecStd     | 4.0    | 0.7   | 0.7   | 0.6   | 1.3   | 1.2   | 1.4   | 1.1   | 1.0   | 1.5   | 1.3   | 1.0   | 0.7   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 37.2  | 40.0  | 45.9  | 57.7  | 66.6  | 74.3  | 77.1  | 76.4  | 70.7  | 61.2  | 52.3  | 42.3  |  |  |
|                                                                        |             | MCWB   | 33.0  | 34.9  | 39.1  | 46.3  | 51.6  | 57.1  | 62.8  | 60.5  | 59.2  | 52.7  | 45.1  | 36.9  |  |  |
|                                                                        | 2%          | DB     | 35.1  | 37.0  | 41.3  | 51.1  | 60.7  | 68.0  | 71.4  | 72.1  | 66.9  | 56.5  | 46.4  | 39.8  |  |  |
|                                                                        |             | MCWB   | 31.7  | 33.2  | 36.1  | 41.2  | 48.6  | 56.7  | 59.9  | 59.8  | 58.1  | 49.7  | 41.4  | 35.5  |  |  |
|                                                                        | 5%          | DB     | 33.7  | 35.3  | 38.2  | 46.8  | 56.9  | 64.2  | 68.4  | 69.2  | 64.5  | 53.9  | 44.1  | 38.2  |  |  |
|                                                                        |             | MCWB   | 30.7  | 32.4  | 33.8  | 38.6  | 47.7  | 54.5  | 59.5  | 59.5  | 57.8  | 48.3  | 39.9  | 34.7  |  |  |
|                                                                        | 10%         | DB     | 31.6  | 33.0  | 35.9  | 43.6  | 53.8  | 61.0  | 65.9  | 66.8  | 62.5  | 51.7  | 42.1  | 36.5  |  |  |
|                                                                        |             | MCWB   | 29.0  | 30.1  | 32.4  | 36.4  | 46.0  | 53.2  | 58.3  | 59.6  | 57.0  | 46.4  | 38.2  | 33.4  |  |  |
|                                                                        | 0.4%        | WB     | 34.1  | 36.5  | 42.1  | 47.4  | 56.4  | 61.6  | 65.5  | 66.6  | 63.2  | 54.5  | 46.4  | 38.3  |  |  |
|                                                                        |             | MCDB   | 36.5  | 38.6  | 44.2  | 55.4  | 62.5  | 68.4  | 69.6  | 70.2  | 67.1  | 58.1  | 50.9  | 41.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 32.5  | 34.1  | 37.2  | 42.6  | 51.9  | 58.7  | 63.2  | 64.1  | 61.3  | 51.2  | 43.0  | 36.5  |  |  |
|                                                                        |             | MCDB   | 34.5  | 36.3  | 39.0  | 49.4  | 57.5  | 64.5  | 68.5  | 67.5  | 64.6  | 55.2  | 46.2  | 38.7  |  |  |
|                                                                        | 5%          | WB     | 30.9  | 32.5  | 34.4  | 39.2  | 49.6  | 56.6  | 61.4  | 62.6  | 59.6  | 48.9  | 40.6  | 35.2  |  |  |
|                                                                        |             | MCDB   | 33.4  | 35.1  | 37.5  | 45.6  | 54.9  | 62.2  | 66.7  | 66.2  | 63.1  | 52.6  | 43.5  | 37.8  |  |  |
|                                                                        | 10%         | WB     | 28.8  | 30.2  | 32.7  | 37.1  | 47.2  | 54.6  | 59.7  | 61.1  | 57.6  | 46.8  | 38.7  | 33.4  |  |  |
|                                                                        |             | MCDB   | 31.8  | 32.9  | 35.5  | 42.2  | 52.5  | 60.3  | 64.8  | 65.2  | 61.3  | 50.4  | 42.0  | 36.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.5  | 13.3  | 10.3  | 9.7   | 12.5  | 14.2  | 11.8  | 8.7   | 7.8   | 7.9   | 8.3   | 11.1  |  |  |
|                                                                        |             | MCDBR  | 11.8  | 11.1  | 10.0  | 15.4  | 18.5  | 20.0  | 15.0  | 11.7  | 9.9   | 11.2  | 9.0   | 7.8   |  |  |
|                                                                        |             | MCWBR  | 11.6  | 9.7   | 7.5   | 9.3   | 11.4  | 12.0  | 8.5   | 6.9   | 7.3   | 9.0   | 8.2   | 7.5   |  |  |
|                                                                        | 5% WB       | MCDBR  | 11.4  | 9.1   | 9.1   | 14.3  | 17.1  | 17.8  | 13.8  | 10.3  | 8.5   | 10.7  | 8.7   | 7.9   |  |  |
|                                                                        |             | MCWBR  | 11.7  | 8.2   | 7.2   | 9.6   | 12.4  | 12.7  | 9.6   | 8.1   | 7.5   | 9.5   | 8.7   | 7.9   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.269 | 0.271 | 0.302 | 0.340 | 0.361 | 0.379 | 0.398 | 0.374 | 0.357 | 0.316 | 0.290 | 0.266 |  |  |
|                                                                        | taud        |        | 2.439 | 2.424 | 2.402 | 2.381 | 2.416 | 2.397 | 2.397 | 2.462 | 2.488 | 2.564 | 2.538 | 2.490 |  |  |
|                                                                        | Ebn at Noon |        | 260   | 286   | 292   | 289   | 285   | 280   | 273   | 275   | 270   | 266   | 251   | 248   |  |  |
|                                                                        | Edh at Noon |        | 20    | 25    | 31    | 35    | 35    | 36    | 36    | 32    | 29    | 23    | 18    | 17    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 338   | 659   | 1075  | 1497  | 1738  | 1872  | 1888  | 1623  | 1165  | 688   | 361   | 273   |  |  |
|                                                                        | RadStd      |        | 40    | 70    | 95    | 154   | 115   | 128   | 102   | 120   | 80    | 86    | 37    | 27    |  |  |

## Historical Trends

|                              |  | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (16 neighbors) |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page